Chapter 8 — Data Communication and Networks

Syllabus (CEN 05/2025, Annexure-VII C, Unit 8): Introduction to data communication; Hardware and interface; Introduction to Networks and Networking devices; Local Area Network and Wide Area Network; Internet working.

8.0 Weightage & Trend Box

Parameter	Detail
Weight in Technical section	**~8% → 4 questions of 50** (scaled: 8 of 100) in the 22-04-2025 CBT-2 paper
Difficulty level observed	Easy–moderate; purely conceptual, **no numericals** appeared, but Nyquist/Shannon and subnetting are standard JE-level askables
Negative marking	−1/3, so guess only when you can eliminate two options

Actual questions observed in the 22-04-2025 CBT-2 (Electronics & Allied) paper from this unit:

1. **Modem vs NIC** — which statement correctly distinguishes a modem from a Network Interface Card.
2. **UTP vs STP** — identify the **incorrect** statement (the trap: "UTP has higher EMI resistance than STP" — FALSE; the shield in STP gives it the higher EMI immunity).
3. **Topology failure behaviour** — how star / bus / ring / mesh topologies behave when a node or cable fails.
4. **SNMP** — the manager–agent model and which messages (GET/SET/TRAP) flow in which direction.

Preparation strategy: This chapter is fact-dense rather than derivation-heavy. Master the comparison tables (topologies, devices vs OSI layer, UTP/STP, TCP/UDP, port numbers) — RRB lifts questions almost verbatim from such tables. Do the 8 worked examples so that a Shannon-capacity or subnetting numerical, if it appears, is free marks.

8.1 Introduction to Data Communication

8.1.1 What is data communication?

Data communication is the exchange of data between two devices via some transmission medium (wire, cable, or air). For communication to occur, the devices must be part of a communication system made up of hardware (physical equipment) and software (programs and protocols).

The effectiveness of a data-communication system depends on four fundamental characteristics:

Characteristic	Meaning
Delivery	Data must reach the correct destination, and only the intended device/user
Accuracy	Data must be delivered without errors (altered bits are unusable)
Timeliness	Data must arrive in time to be useful (critical for audio/video — "real-time" delivery)
Jitter	Variation in packet arrival time; uneven delay ruins audio/video quality

8.1.2 Components of a data-communication system (five components)

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Sender → [Message] → Transmission medium → Receiver
              ↑ governed by Protocol (both ends) ↑
```

#	Component	Role	Example
1	**Message**	The information to be communicated	Text, numbers, pictures, audio, video
2	**Sender**	Device that sends the message	Computer, phone, workstation, camera
3	**Receiver**	Device that receives the message	Computer, phone, TV
4	**Transmission medium**	Physical path from sender to receiver	Twisted pair, coax, fibre, radio waves
5	**Protocol**	Set of rules governing communication — an *agreement* between communicating devices	TCP/IP, HTTP, Ethernet

> Without a protocol, two connected devices may be linked but cannot *communicate* — like two people speaking different languages.

8.1.3 Data representation

Data type	Representation
Text	Bit patterns — coded character sets: **ASCII** (7-bit, 128 symbols), extended ASCII (8-bit), **Unicode** (32-bit, all world scripts; ASCII = first 127 Unicode characters)
Numbers	Binary directly (not character codes) for arithmetic
Images	Matrix of **pixels**; each pixel a bit pattern (RGB / YCM colour models)
Audio	Continuous signal sampled and digitised (PCM)
Video	Sequence of images (frames) + audio

8.1.4 Data flow (direction of transmission) — **frequent one-mark question**

Mode	Direction	Description	Example
Simplex	One way only	One device transmits, the other only receives; entire channel capacity in one direction	Keyboard → CPU, TV/radio broadcast
Half-duplex	Both ways, **one at a time**	Each station can both transmit and receive, but not simultaneously; entire capacity taken by whichever is sending	Walkie-talkie, CB radio
Full-duplex (duplex)	Both ways **simultaneously**	Either two physical paths, or channel capacity divided between the two directions	Telephone network, modern switched Ethernet

8.1.5 Serial vs parallel transmission

Feature	Parallel	Serial
Bits sent	n bits at a time over n wires	1 bit at a time over a single line
Speed (same clock)	n× faster	Slower per clock, but clocks can be much higher
Cost & distance	Expensive; crosstalk/skew limits to short distances (inside PC, old printer port)	Cheap; long distances (USB, RS-232, SATA, Ethernet, PCIe lanes)
Conversion	—	Needs parallel↔serial converters (UART/shift registers) at the interfaces

8.1.6 Synchronous vs asynchronous serial transmission

Feature	Asynchronous	Synchronous
Timing	No common clock; timing recovered per character	Common/embedded clock; continuous bit stream
Framing	Each byte framed by **1 start bit (0)** and **1–2 stop bits (1)**; gaps allowed between bytes	Data sent as long frames/blocks; no start–stop bits per byte
Overhead	High (≥ 2 extra bits per 8 data bits ≈ 20%+)	Low — higher efficiency
Speed	Lower	Higher
Example	RS-232 terminal, keyboard (UART)	HDLC, Ethernet frames, SONET

*(A third class, **isochronous**, guarantees data arrives at a fixed rate — used for real-time audio/video, e.g. USB isochronous transfers.)*

8.1.7 Bit rate vs baud rate — **classic trap**

- **Bit rate** = number of **bits** transmitted per second (bps).
- **Baud rate** = number of **signal units (symbols)** transmitted per second.
- If each signal element carries r bits: **Bit rate = Baud rate × r**, where r = $\log_2 L$ (L = number of signal levels).
- Baud rate ≤ bit rate always (equal when 1 bit/symbol). **Bandwidth required is proportional to baud rate, not bit rate** — that is why multi-level modulation (QAM) packs high bit rates into limited bandwidth.

Quick example: A 16-QAM signal (L = 16 → 4 bits/symbol) at 2400 baud carries 2400 × 4 = **9600 bps**.

8.1.8 Bandwidth, throughput, latency

Term	Meaning
Bandwidth (Hz)	Range of frequencies a channel passes (analog sense)
Bandwidth (bps)	Maximum data rate a link can carry (digital, loose usage)
Throughput	*Actual* measured rate of successful data delivery — always ≤ bandwidth
Latency (delay)	Time for a complete message to travel source → destination = propagation time + transmission time + queuing time + processing delay
Propagation time	Distance ÷ propagation speed (~2×10^8 m/s in cable/fibre)
Transmission time	Message size ÷ bandwidth
Bandwidth–delay product	Number of bits that can "fill the link" = bandwidth × round-trip delay

8.1.9 Channel capacity — Nyquist and Shannon (formulas + when to use which)

Nyquist bit rate (noiseless channel):

> **$C = 2 \times B \times \log_2 L$** bits/second

where B = bandwidth (Hz), L = number of discrete signal levels. Increasing L raises capacity but makes the receiver's job harder (levels closer together).

Shannon capacity (noisy channel) — theoretical upper limit:

> **$C = B \times \log_2 (1 + SNR)$** bits/second

where SNR is the **linear** (not dB) signal-to-noise ratio. Convert: SNR(linear) = 10^(SNR_dB/10). Handy identity: **$SNR_{dB} = 10 \log_{10} SNR$**.

- Shannon gives the *ceiling* no coding scheme can beat; Nyquist tells you *how many levels* you need to approach it.
- In practice: compute Shannon C for the limit, then choose a working rate below it and use Nyquist to find L.

(Worked Examples 1–3 in §8.10.)

8.2 Transmission Media

Transmission media sit **below the Physical layer** of the OSI model. Two broad classes:

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Transmission media
├── Guided (wired): twisted pair, coaxial, optical fibre
└── Unguided (wireless): radio, microwave, infrared, satellite
```

8.2.1 Twisted-pair cable

Two insulated copper conductors twisted together. **Twisting cancels externally induced noise**: interference affects both wires nearly equally, so the *difference* signal (which carries the data) stays clean; more twists per unit length → better noise rejection.

UTP vs STP — 2025 PYQ area:

Feature	UTP (Unshielded Twisted Pair)	STP (Shielded Twisted Pair)
Construction	Twisted pairs in plastic jacket only	Each pair (or the bundle) wrapped in **metal foil/braid shield**
EMI / noise immunity	Lower — relies on twisting alone	**Higher — shield blocks electromagnetic interference**
Cost	Cheaper	Costlier
Installation	Easy, flexible, thin	Bulkier; shield must be **grounded** to work
Speed/attenuation	Comparable per category; STP performs better in electrically noisy environments	—
Usage	Overwhelmingly common in LANs, telephone	Industrial plants, near motors/heavy machinery

> ⚠ **PYQ trap (22-04-2025):** "UTP has higher resistance to EMI than STP" is **FALSE**. STP's grounded shield gives it superior EMI resistance. UTP's advantages are *cost, flexibility and ease of installation*, never EMI immunity.

UTP categories (EIA/TIA-568) — memorise Cat 5e/6/7:

Category	Max data rate	Bandwidth	Typical use
Cat 3	10 Mbps	16 MHz	Old telephone / 10Base-T
Cat 5	100 Mbps	100 MHz	Fast Ethernet
Cat 5e	**1 Gbps**	100 MHz	Gigabit Ethernet (most common legacy LAN cable)
Cat 6	1 Gbps (10 Gbps up to ~55 m)	250 MHz	Modern LAN standard
Cat 6a	10 Gbps (100 m)	500 MHz	10G LANs
Cat 7	10 Gbps	600 MHz	**Shielded (S/FTP) by design**; data centres
Cat 8	25/40 Gbps	2000 MHz	Short data-centre links (~30 m)

Standard UTP connector: **RJ-45** (8 pins). Max segment length for LAN Ethernet over UTP: **100 m**.

8.2.2 Coaxial cable

- Central copper core → insulating dielectric → braided outer conductor (acts as **shield + return path**) → plastic jacket.
- Carries higher-frequency signals than twisted pair; good noise immunity because of concentric shield; but higher attenuation → needs repeaters/amplifiers for long runs.
- Connector: **BNC** (also T-connector, terminator for old bus LANs).
- RG ratings: **RG-59** (75 Ω, cable TV), **RG-58** (50 Ω, thin Ethernet 10Base2), RG-11 (thick Ethernet 10Base5).
- Uses: cable TV networks, traditional Ethernet buses, CCTV.

8.2.3 Optical fibre

Transmits data as **light pulses** through a glass/plastic core using **total internal reflection** (core refractive index > cladding refractive index).

Item	Detail
Structure	Core (n_1) → cladding ($n_2 < n_1$) → buffer/jacket
Propagation modes	**Multimode step-index**, **multimode graded-index**, **single-mode**
Single-mode	Tiny core (~8–10 μm), laser source, longest distance, highest bandwidth, costliest
Multimode	Larger core (50/62.5 μm), LED source, shorter links (buildings/campus)
Sources / detectors	LED or laser diode / photodiode (PIN, APD)
Connectors	SC, ST, LC, MT-RJ

Advantages: enormous bandwidth (THz), very low attenuation (repeaters tens of km apart), **complete immunity to EMI** (light, not electricity), no crosstalk, light weight, hard to tap (secure), no spark hazard.

Disadvantages: costly installation and skilled splicing, unidirectional per fibre (pairs used), fragile to sharp bends.

Guided media comparison (exam table):

Feature	Twisted pair	Coaxial	Optical fibre
Bandwidth / data rate	Low–moderate	Moderate–high	**Highest**
EMI immunity	UTP low, STP moderate	Good	**Complete**
Attenuation	High	Moderate	**Lowest**
Cost	**Lowest**	Moderate	Highest
Distance per segment	~100 m	~500 m	km–tens of km
Installation	Easiest	Moderate	Hardest (splicing)

8.2.4 Unguided (wireless) media

Signals travel through free space; **antennas** radiate/receive. Propagation: ground wave (< 2 MHz), sky wave / ionospheric (2–30 MHz), line-of-sight (> 30 MHz).

Medium	Band (typical)	Character	Uses
Radio waves	3 kHz – 1 GHz	**Omnidirectional**, penetrate walls, prone to interference; licensed bands	AM/FM radio, TV, cordless phones, paging
Microwaves	1 – 300 GHz	**Unidirectional, line-of-sight**; parabolic dish/horn antennas; towers ~50 km apart (curvature); do not penetrate walls well; rain attenuation at high f	Terrestrial point-to-point links, cellular backhaul, satellite links, Wi-Fi (2.4/5 GHz ISM)
Infrared	300 GHz – 400 THz	Very short range, **cannot penetrate walls** (advantage: no inter-room interference; disadvantage: same-room only); unusable outdoors (sunlight)	TV remotes, IrDA device-to-device
Satellite	C (4/6 GHz), Ku (11/14), Ka bands	GEO satellite at **35,786 km** acts as repeater (transponder); one-hop covers ⅓ of Earth; **propagation delay ~270 ms** up+down	DTH TV, VSAT, GPS (MEO), remote-area links

8.3 Encoding, Modems and the NIC

8.3.1 Line coding one-liners (digital data → digital signal)

Scheme	Rule	Key point
Unipolar NRZ	1 = +V, 0 = 0 V	DC component; sync loss on long runs of same bit
Polar NRZ-L	1 = one level, 0 = the other	Level itself is the bit
NRZ-I	1 = **invert** at bit start, 0 = no change	Transition = 1 (differential; used in USB with stuffing)
RZ (Return-to-Zero)	Signal returns to 0 mid-bit	Self-syncing but needs **double bandwidth**
Manchester	**Transition in the middle of every bit** (low→high = 1, high→low = 0 per IEEE 802.3)	Self-clocking, no DC; used in **10 Mbps Ethernet**; needs 2× bandwidth
Differential Manchester	Mid-bit transition always (clock); transition **at start** = 0	Used in Token Ring
Bipolar AMI	0 = zero, 1 = alternating ±V	No DC; used in T1 lines

8.3.2 Modem — MOdulator/DEModulator

- Converts the computer's **digital** data into an **analog** signal suitable for an analog channel (telephone local loop) — *modulation* — and converts received analog back to digital — *demodulation*.
- Works with ASK/FSK/PSK/QAM constellations; V-series ITU standards (V.32 = 9600 bps, V.90/V.92 = 56 kbps down).
- Modem is a **DCE** (Data Circuit-terminating Equipment); computer/terminal is the **DTE**.
- Modern usage: DSL modem (digital data over telephone copper at higher frequencies), cable modem (DOCSIS over coax), optical modem/ONT.
- **56k asymmetric note:** V.90 gives 56 kbps downstream / 33.6 kbps upstream because the ISP side is digitally connected.

8.3.3 NIC (Network Interface Card) and MAC address

- The NIC (network adapter/LAN card) is the hardware that connects a host to the network medium. It frames data, computes the frame check sequence, performs media access (CSMA/CD for classic Ethernet), and does the parallel↔serial conversion onto the wire.
- Operates at the **Physical + Data Link layers (Layers 1–2)** of OSI.
- Every NIC has a globally unique **MAC address: 48 bits (6 bytes)**, written as 12 hex digits, e.g. `00:1A:2B:3C:4D:5E`. First 3 bytes = **OUI** (manufacturer ID assigned by IEEE); last 3 bytes = device serial. MAC address is **burned into ROM** (hence "physical/hardware/burned-in address"). Broadcast MAC = `FF:FF:FF:FF:FF:FF`.

8.3.4 Modem vs NIC — **2025 PYQ comparison table**

Point	Modem	NIC
Core job	**Signal conversion:** digital ↔ analog (modulation/demodulation) to use an analog channel (phone line/coax/RF)	**Network attachment:** physical + data-link interface of a host to a (digital) LAN
Typical use	Connects a computer/site to the **internet/WAN via the telephone or cable network**	Connects a computer to a **LAN (Ethernet/Wi-Fi)**
Conversion	Digital data → analog signal and back	Data → digital line-coded frames (no D/A of the modem type)
OSI position	Physical layer device (DCE)	Layers 1–2; owns the **MAC address**
Addressing	No host address of its own on the LAN	48-bit MAC address
Form	External/internal box between DTE and line	Card/chip on the motherboard

> **Exam answer:** "A modem converts digital signals to analog (and vice-versa) for transmission over telephone lines, whereas a NIC provides the physical interface between a computer and the network and has a unique MAC address." Both are needed in a home setup: NIC → router → modem → ISP.

8.4 Hardware and Interfaces

8.4.1 DTE and DCE

- **DTE (Data Terminal Equipment):** source/sink of digital data — computer, terminal, printer.
- **DCE (Data Circuit-terminating Equipment):** sits between DTE and the transmission line; converts/conditions signals — modem, NT1, CSU/DSU.
- Interface standards (RS-232 etc.) define the **DTE–DCE boundary**: mechanical (connector), electrical (voltage levels), functional (pin meanings) and procedural (handshake sequence) characteristics.

8.4.2 RS-232 (EIA-232) — the classic serial interface

Item	Detail
Type	**Asynchronous serial**, point-to-point, DTE↔DCE
Connector	DB-25 (original) / **DB-9** (common)
Logic levels	**Logic 1 (mark) = −3 to −15 V; Logic 0 (space) = +3 to +15 V** (negative logic — a favourite trick question)
Max cable length / speed	~15 m (50 ft) at 20 kbps nominal (practically 115.2 kbps short runs)
Mode	Full-duplex (separate TxD, RxD)
Key DB-9 pins	2 RxD, 3 TxD, 5 GND (signal ground), 7 RTS, 8 CTS, 4 DTR, 6 DSR, 1 DCD, 9 RI
Handshake	Hardware (RTS/CTS) or software (XON/XOFF)
Null modem	Cross-connected cable (TxD↔RxD) to link two DTEs directly

8.4.3 USB (Universal Serial Bus)

- Serial, hot-pluggable, plug-and-play; **tiered-star topology** with one host controller; up to **127 devices** per host; supplies power (5 V).
- Differential pair D+/D− (USB 2.0), NRZI encoding with bit stuffing.

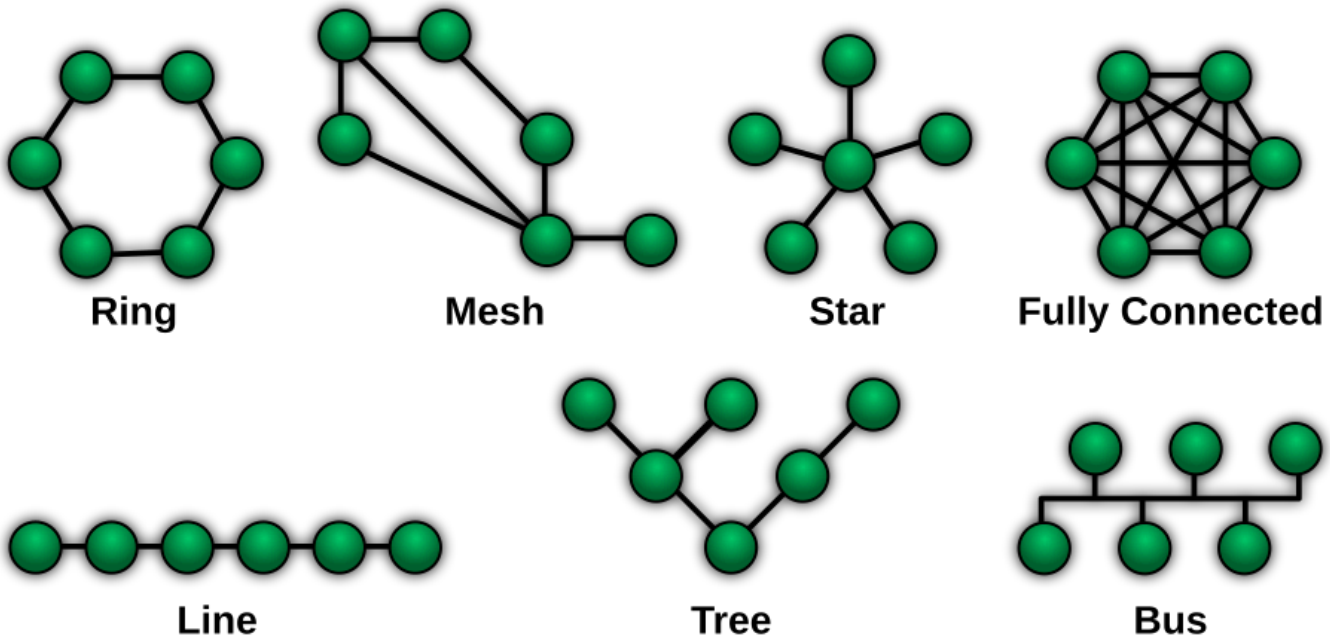
Version	Speed	Marketing name
USB 1.1	1.5 Mbps (low) / 12 Mbps (full)	—
USB 2.0	**480 Mbps**	Hi-Speed
USB 3.0 / 3.1 Gen1	5 Gbps	SuperSpeed
USB 3.1 Gen2	10 Gbps	SuperSpeed+
USB4	20–40 Gbps	—

- Transfer types: control, bulk, interrupt, **isochronous** (guaranteed rate for audio/video).
- Connectors: Type-A, Type-B, Mini, Micro, reversible **Type-C**.

8.4.4 RJ-45 and Ethernet cabling basics

- **RJ-45**: 8-position 8-contact (8P8C) modular connector terminating 4 twisted pairs of UTP.
- Wiring standards **T568A** and **T568B** (differ only in swapping green and orange pairs).
- **Straight-through cable** (same standard both ends): PC → switch/hub (unlike devices).
- **Crossover cable** (T568A one end, T568B other): PC → PC, switch → switch (like devices) — largely obsolete because modern ports do **Auto-MDIX**.
- 10/100 Mbps Ethernet uses pins **1,2 (transmit)** and **3,6 (receive)**; Gigabit uses all 8 pins/4 pairs bidirectionally.
- RJ-11 (6P2C/6P4C) is the smaller **telephone** connector — don't confuse.

8.5 Networks: Types and Topologies



Bus, star, ring, mesh and tree topologies.

8.5.1 What is a network? Criteria

A **network** = a set of devices (nodes) connected by communication links to share resources (data, printers, internet). Key criteria: **performance** (throughput, delay), **reliability** (failure frequency, recovery time, robustness), **security**.

Connection types: point-to-point (dedicated link between two devices) vs **multipoint/broadcast** (link shared by many devices).

8.5.2 Network categories by geographic scale

Type	Full form	Span	Ownership	Examples
PAN	Personal Area Network	~10 m, around a person	Individual	Bluetooth, USB tether
LAN	Local Area Network	Room–building–campus (≤ a few km)	Single organisation (private)	Office Ethernet, home Wi-Fi
MAN	Metropolitan Area Network	A city (~5–50 km)	Single or multiple operators	Cable-TV network, city Wi-Fi/WiMAX
WAN	Wide Area Network	Country / world	Usually leased from carriers	Internet, bank branch network, leased lines

- LAN: **high speed (100 Mbps–10 Gbps), low error rate, cheap, privately owned**.
- WAN: lower speed per cost, higher delay, uses public/leased carrier infrastructure, connects LANs via **routers**.
- **Internetwork (internet, small i)** = two or more networks connected; **the Internet (capital I)** = the specific global internetwork.

8.5.3 Network topologies — **the biggest PYQ magnet in this unit**

Topology = the geometric layout of links and nodes.

Bus topology

- One **shared backbone cable**; every node taps on via drop lines; both ends terminated (terminators absorb signals to stop reflections).
- Cheapest cabling (least cable for small nets); used in legacy 10Base2/10Base5 Ethernet.
- **Node failure:** network keeps working (a dead station simply stops talking).
- **Backbone-cable failure/break: entire network goes down** (and reflections from the break disrupt both sides).
- Other cons: difficult fault isolation, collisions on the shared medium, hard to add nodes mid-run, degraded performance under heavy load.

Star topology

- Every node connects by a **dedicated point-to-point link to a central device** (hub/switch).
- Modern Ethernet LAN standard layout.
- **Node or single-cable failure: only that node is cut off — rest of network unaffected.** Easiest fault detection and isolation.
- **Central device (hub/switch) failure: whole network goes down — single point of failure.**
- More cable than bus (one run per node) but cheap installation & easy expansion.

Ring topology

- Each node connects to exactly **two neighbours**, forming a closed loop; data circulates (usually unidirectionally) node-to-node, each node **regenerating** the signal; access controlled by **token passing** → no collisions.
- **One node or one link failure breaks the ring — can disable the whole network** (unless a dual/counter-rotating ring, e.g. FDDI, provides wrap-around, or nodes are bypassed by relays).
- Adding/removing a node disturbs the whole ring. Predictable (deterministic) performance under load.

Mesh topology

- **Every node has a dedicated point-to-point link to every other node** (full mesh).
- Number of duplex links for n nodes: **$n(n-1)/2$**; each node needs **$n-1$ ports**.
- **Most robust/fault-tolerant: failure of any one link or node does NOT bring the network down — traffic reroutes** via alternative paths. Also most secure/private (dedicated links) and no traffic contention.
- **Most expensive** in cable and ports; installation/maintenance hardest. Used where reliability is paramount (backbones, military); *partial mesh* is the practical compromise.

Tree (hierarchical) topology

- **Star networks connected to a central bus/backbone** (star-bus hybrid); root hub → secondary hubs → nodes.
- Scalable, easy expansion & fault isolation per branch; **backbone failure splits the network**; root failure severs branches.

Hybrid topology

- Any combination (e.g. star-ring, star-bus) inheriting pros/cons of its parts; virtually all real large networks are hybrids.

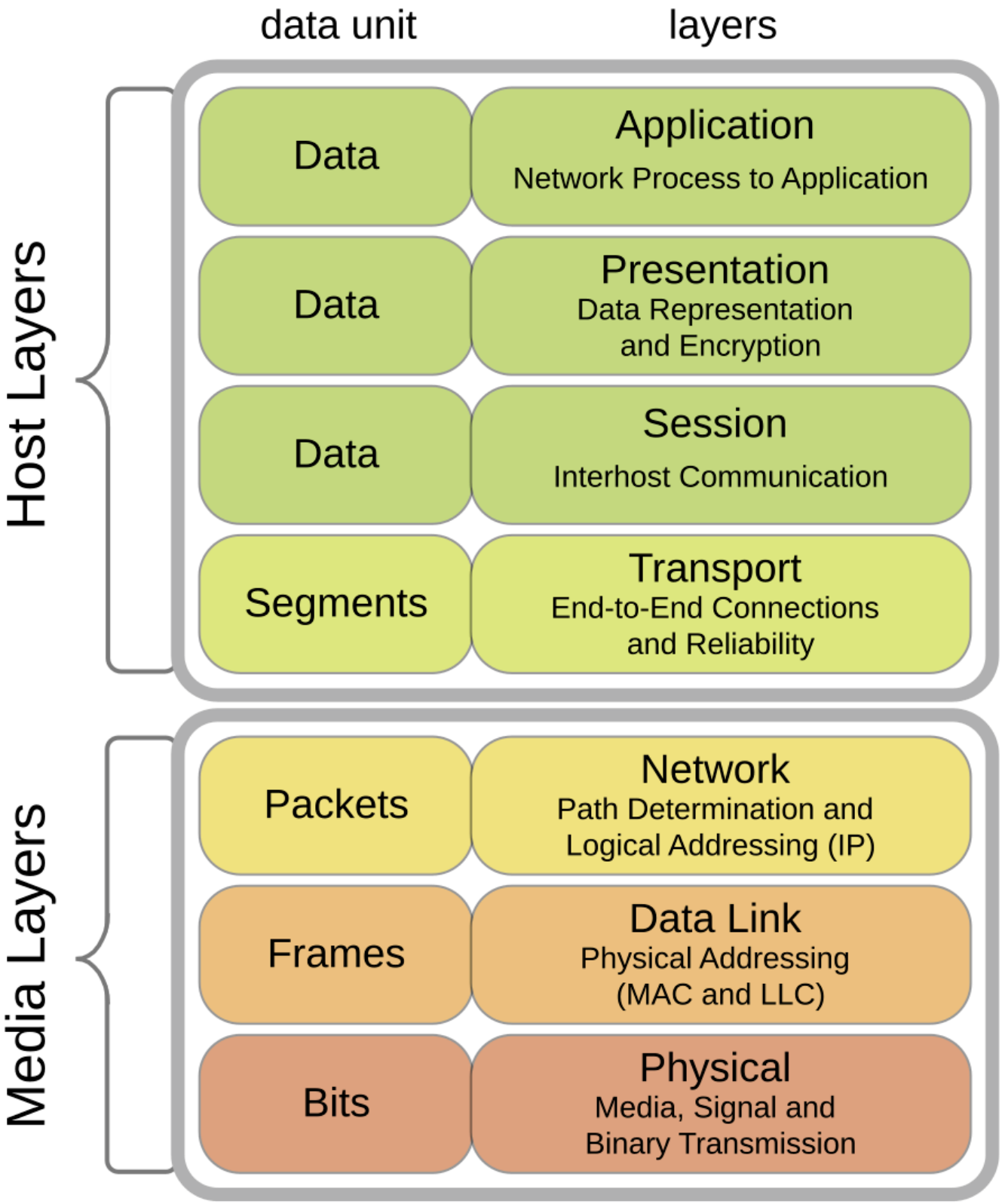
Topology master comparison table ☆

Topology	Cable cost	Node failure effect	Cable/link failure effect	Central point of failure	Ease of expansion	Access method / notes
Bus	Lowest	Others unaffected	**Backbone break kills entire network**	Backbone itself	Poor	CSMA/CD; terminators needed
Star	Moderate	**Only that node lost**	Only that node lost	**Yes — hub/switch**	**Best**	Modern Ethernet; easiest troubleshooting
Ring	Moderate	**Can break whole ring**	**Breaks whole ring** (single ring)	No single centre, every node critical	Poor (disturbs ring)	Token passing; no collisions
Mesh	**Highest** — n(n−1)/2 links	**Network survives (reroutes)**	**Network survives**	**None**	Hard (n−1 new links per node)	Most reliable & secure
Tree	High	Branch above it affected	Backbone break partitions network	Root/backbone	Good	Hierarchical; DSL/cable-TV nets

> **2025 PYQ answer key:** star → one node failing does not affect others but hub failure kills all; bus → cable break kills all; ring → any single break kills all (single ring); mesh → survives individual failures.

8.6 Network Reference Models

8.6.1 OSI model (ISO — Open Systems Interconnection, 7 layers)



OSI seven-layer reference model.

Mnemonics: bottom-up "**P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way"; top-down "**A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing".

#	Layer	Core functions	PDU name	Addressing	Typical devices/protocols
7	**Application**	Network services to user apps: file transfer, mail, remote login, directory	Data (message)	—	HTTP, FTP, SMTP, DNS, SNMP, Telnet
6	**Presentation**	Translation (syntax/EBCDIC↔ASCII), **encryption/decryption, compression**	Data	—	SSL/TLS (conceptually), JPEG, MPEG
5	**Session**	Dialog control (establish/maintain/terminate sessions), **synchronisation (checkpoints)**	Data	—	NetBIOS, RPC, sockets API
4	**Transport**	**Process-to-process (end-to-end) delivery**, segmentation & reassembly, connection control, end-to-end flow & error control	**Segment** (TCP) / Datagram (UDP)	**Port number**	TCP, UDP; (firewalls filter here)
3	**Network**	**Source-to-destination (host-to-host) delivery across networks: logical addressing & routing**	**Packet**	**IP address**	**Router**, L3 switch; IP, ICMP, ARP*, routing protocols
2	**Data Link**	**Node-to-node (hop-to-hop) delivery**: framing, physical (MAC) addressing, per-hop flow/error control (CRC), media access (MAC sublayer + LLC sublayer)	**Frame**	**MAC address**	**Switch, bridge, NIC**; Ethernet, HDLC, PPP
1	**Physical**	Transmit **raw bits**: media, connectors, voltages, bit rate, line encoding, topology, transmission mode	**Bit**	—	**Hub, repeater, cables, modem**

Key one-liners the exam loves:

- **Routing → Network layer. Framing & MAC address → Data Link. Port numbers & end-to-end reliability → Transport. Encryption/compression → Presentation. Dialog/synchronisation → Session.**

- Data Link = hop-to-hop; Network = host-to-host (source-to-destination); Transport = process-to-process.
- Encapsulation going down (each layer adds its header; Data Link also adds a trailer); decapsulation going up.

8.6.2 TCP/IP model and mapping

TCP/IP (4-layer, RFC view)	5-layer teaching model	Maps to OSI	Protocols
Application	Application	Application + Presentation + Session (5–7)	HTTP, FTP, SMTP, DNS, SNMP
Transport	Transport	Transport (4)	TCP, UDP, SCTP
Internet	Network	Network (3)	IP, ICMP, IGMP, ARP
Network Access (Host-to-Network / Link)	Data Link + Physical	Layers 1–2	Ethernet, Wi-Fi, PPP

OSI vs TCP/IP: OSI is a *prescriptive reference model* (protocol-independent, defined before its protocols); TCP/IP is a *descriptive, implemented* suite (protocols came first). OSI separates Presentation & Session; TCP/IP folds them into Application. The Internet runs on TCP/IP.

8.7 Networking Devices — which layer, which domain ⭐

Device	OSI layer	Function	Collision domains	Broadcast domains
Passive hub	— (below L1, just wiring)	Connection point only; **no regeneration**	one shared	one
Repeater	**1 – Physical**	**Regenerates** (not merely amplifies) the digital signal to extend segment length; 2 ports; cannot filter or connect different LAN types	Does **not** segment — one big collision domain	One
Active hub	**1 – Physical**	Multiport repeater; broadcasts every bit out of all ports; half-duplex	**One collision domain for all ports**	One
Bridge	**2 – Data Link**	Connects two LAN segments; **filters/forwards frames by MAC address** using a learned table; regenerates too	**Each port = separate collision domain**	One
Switch	**2 – Data Link** (L3 switches exist)	Multiport bridge; **MAC (CAM) table**; forwards a frame only to the destination port (unicast); full-duplex, dedicated bandwidth per port; methods: store-and-forward / cut-through	**Every port its own collision domain**	One (all ports, unless VLANs)
Router	**3 – Network**	Connects **different networks**; forwards **packets by IP address** using a routing table; chooses best path; **blocks broadcasts** by default; can join dissimilar link technologies	Each port separate	**Each port a separate broadcast domain**
Gateway	Up to **7 – Application** (protocol converter)	Connects networks using **entirely different protocol architectures**; translates protocols/data formats; "default gateway" in IP configs = the router to other networks	—	Separates
Access Point (AP)	2	Bridges wireless 802.11 stations to the wired LAN	Wireless is shared medium	One
Firewall	3–4 (packet filter) up to 7 (application/proxy)	Enforces security policy on traffic entering/leaving the network	—	—
Modem	1	Digital↔analog conversion for WAN	—	—

Device	OSI layer	Function	Collision domains	Broadcast domains
		access		
NIC	1–2	Host's network interface; MAC address	—	—

Hub vs Switch (asked repeatedly):

Point	Hub	Switch
Layer	Physical (1)	Data Link (2)
Forwarding	Broadcasts to **all** ports	Only to destination port (**MAC table**)
Collision domain	Single, shared	Per-port (collisions eliminated with full duplex)
Duplex / bandwidth	Half-duplex, shared bandwidth	Full-duplex, dedicated per port
Intelligence / security	None (dumb)	Learns addresses; better security & performance

Collision domain = set of devices whose simultaneous transmissions can collide (segmented by bridge/switch/router).

Broadcast domain = set of devices that receive each other's L2 broadcasts (segmented **only by a router** or VLANs).

8.8 LAN Technologies

8.8.1 Ethernet (IEEE 802.3)

Dominant LAN technology, invented at Xerox PARC (Metcalfe), standardised as **IEEE 802.3**.

Generation	Standard names	Speed	Media	Notes
Standard Ethernet	10Base5 (thick coax), 10Base2 (thin coax), **10Base-T** (UTP Cat3, star-hub), 10Base-F	**10 Mbps**	coax / UTP / fibre	Manchester encoding; CSMA/CD; half-duplex
Fast Ethernet (802.3u)	**100Base-TX** (2 pairs Cat5), 100Base-FX (fibre), 100Base-T4	**100 Mbps**	UTP/fibre	Auto-negotiation introduced
Gigabit Ethernet (802.3z/ab)	**1000Base-T** (4 pairs Cat5e), 1000Base-SX/LX (fibre)	**1 Gbps**	UTP/fibre	Full-duplex switching normal; CSMA/CD practically retired
10-Gigabit (802.3ae/an)	10GBase-T (Cat6a), -SR/-LR	10 Gbps	UTP/fibre	Full-duplex only — **no CSMA/CD at all**

Naming: `<speed>Base<medium>` — "Base" = baseband signalling; T = twisted pair, F/S/L = fibre; 5/2 = old coax max segment (500 m/185 m). Max UTP segment: **100 m**; max nodes per legacy segment rules ("5-4-3": 5 segments, 4 repeaters, 3 populated).

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) — the access method of classic (half-duplex, shared) Ethernet:

1. **Carrier sense:** listen; transmit only if the medium is idle.
2. **Transmit and keep listening** while sending.
3. **Collision detected** (received ≠ sent / voltage anomaly): abort, send a **32-bit jam signal** so all stations notice.

4. **Back-off:** wait a random time chosen by **binary exponential back-off** (after nth collision pick a random slot from 0…2^n-1, cap n = 10, give up after 16 attempts), then retry.

- Constraint: minimum frame length must outlast twice the end-to-end propagation delay (**slot time**) so the sender is still transmitting when a far-end collision returns → Ethernet **minimum frame = 64 bytes**; frames shorter are padded; a fragment < 64 B is a collision "runt".
- Wireless uses **CSMA/CA (Collision Avoidance)** instead, because a radio cannot transmit and listen for its own collision simultaneously (and due to the hidden-station problem).

Ethernet (802.3) frame basics:

Field	Size
Preamble (7 B, 10101010…) + SFD (1 B, 10101011)	8 B (physical-layer added)
Destination MAC	6 B
Source MAC	6 B
Length/Type	2 B
Data + pad	**46 – 1500 B** (1500 = max payload, MTU)
FCS — CRC-32	4 B
Frame size (excl. preamble)	**64 – 1518 B**

8.8.2 Wi-Fi — IEEE 802.11 family ☆

Wireless LAN; access method **CSMA/CA**; infrastructure mode (via AP, forming a BSS/ESS) or ad-hoc (IBSS); network name = **SSID**; security: WEP (broken) → WPA → **WPA2 (AES-CCMP)** → WPA3.

Standard	Year	Band	Max PHY rate	Note
802.11a	1999	**5 GHz**	54 Mbps	OFDM
802.11b	1999	**2.4 GHz**	11 Mbps	DSSS; longest-lived legacy
802.11g	2003	2.4 GHz	54 Mbps	OFDM, b-compatible
802.11n (Wi-Fi 4)	2009	2.4 & 5 GHz	600 Mbps	**MIMO**, 40 MHz channels
802.11ac (Wi-Fi 5)	2013	5 GHz	~6.9 Gbps	MU-MIMO, 160 MHz, 256-QAM
802.11ax (Wi-Fi 6/6E)	2019	2.4/5 (+6 GHz for 6E)	~9.6 Gbps	OFDMA, 1024-QAM, high density

Related IEEE 802 standards: **802.3 Ethernet, 802.11 WLAN, 802.15 Bluetooth/WPAN, 802.16 WiMAX**, 802.1Q VLAN tagging, 802.1D spanning tree.

8.8.3 Token Ring (IEEE 802.5) — one paragraph

IBM's LAN standard (4/16 Mbps): stations form a logical ring (physically often a star through a MAU — Multistation Access Unit); a small frame called the **token** circulates, and only the station **holding the token may transmit**, so collisions are impossible and access is **deterministic** (bounded worst-case wait — good for real-time/industrial loads). The frame travels around the ring, is copied by the destination, and is removed by the sender, which then releases the token. Uses differential Manchester encoding. FDDI extended the idea to a 100 Mbps dual counter-rotating fibre ring (self-healing on a single break). Both are now obsolete, displaced by switched Ethernet, but token passing remains a favourite exam contrast to CSMA/CD.

8.9 WAN Technologies

8.9.1 Switching techniques

Circuit switching

- A **dedicated physical path** is established end-to-end **before** data flows (setup phase), held for the whole session, then released (teardown).
- Fixed bandwidth reserved whether used or not → inefficient for bursty data, but **no congestion once connected, constant delay, guaranteed rate** — ideal for voice.
- Example: **PSTN telephone network**. Three phases: setup → data transfer → teardown.

Packet switching

- Message broken into **packets**, each carrying addressing; links shared by many flows (**statistical multiplexing**); **store-and-forward** at each node.
- Efficient for bursty computer data; per-packet delays vary (queuing), congestion possible.
- Two flavours:

Feature	Datagram approach	Virtual circuit approach
Setup	None — each packet routed **independently**	Logical path (VC) set up first; all packets follow it
Header carries	**Full destination address**	Short **VC identifier (label)**
Ordering	Packets may arrive **out of order**, may take different routes	**In-order** delivery along fixed path
On node failure	Later packets route around	VC broken; must be re-established
Analogy / examples	Postal letters; **IP, UDP**	Phone-call-like; **X.25, Frame Relay, ATM, MPLS LSPs**

(Message switching = store-and-forward of whole messages — historical telegraph, high storage/delay.)

8.9.2 WAN access technologies — one-liners

Technology	One-liner
PSTN / dial-up	Public Switched Telephone Network; analog local loop + modem; ≤ 56 kbps; circuit-switched
ISDN	Integrated Services Digital Network — end-to-end **digital** telephony; **BRI = 2B + D** (2×64 kbps bearer + 16 kbps signalling = 144 kbps), **PRI = 30B + D** (E1, India/Europe) or 23B + D (T1)
Leased line	Dedicated point-to-point circuit rented from carrier, fixed bandwidth 24×7 (64 kbps…E1 = 2.048 Mbps, T1 = 1.544 Mbps…); secure, costly
DSL / ADSL	Digital data on existing telephone copper above the voice band; **A**symmetric = downstream ≫ upstream; splitter/filter separates voice
Frame Relay	Packet-switched virtual-circuit WAN service (successor of X.25, less error checking, faster); PVCs identified by DLCI
ATM	Asynchronous Transfer Mode — fixed **53-byte cells (48 data + 5 header)**, virtual circuits, QoS classes; once the B-ISDN backbone
MPLS	Multi-Protocol Label Switching — routers forward by short **labels** instead of IP lookup along Label-Switched Paths; "layer 2.5"; backbone of modern carrier VPNs & traffic engineering
SONET/SDH	Optical carrier TDM hierarchy for long-haul trunks (STM-1 = 155.52 Mbps)

8.10 Internetworking and the TCP/IP Suite

Internetworking = connecting multiple, possibly heterogeneous networks with **routers/gateways** so they function as one logical network. The glue is the **Internet Protocol (IP)**: every host gets a logical IP address, and routers forward packets between networks using routing tables (protocols: RIP — hop-count distance vector; OSPF — link state; BGP — inter-AS path vector).

8.10.1 IPv4 addressing

- **32-bit** logical address, written in **dotted decimal**: four octets 0–255, e.g. `192.168.10.5`.
- Total address space = 2^{32} ≈ 4.29 billion.
- Two parts: **Network ID + Host ID** (boundary set by class or by subnet mask/CIDR).

Classful addressing (identify class by the FIRST octet):

Class	First bits	First-octet range	Default mask	Net/Host octets	Networks × hosts	Use
A	0	**1 – 126**	255.0.0.0 (/8)	N.H.H.H	126 nets × 16,777,214 hosts	Very large orgs
B	10	**128 – 191**	255.255.0.0 (/16)	N.N.H.H	16,384 × 65,534	Large orgs
C	110	**192 – 223**	255.255.255.0 (/24)	N.N.N.H	2,097,152 × **254**	Small networks
D	1110	**224 – 239**	—	—	—	**Multicast**
E	1111	**240 – 255**	—	—	—	**Experimental/reserved**

Special addresses:

- **127.x.x.x — loopback** (`127.0.0.1` = localhost); that's why 127 is missing from Class A's usable range.
- Host bits all 0 = **network address**; host bits all 1 = **directed broadcast**; `255.255.255.255` = limited broadcast; `0.0.0.0` = "this host/any".
- Hosts per subnet = 2^(host bits) − 2 (network & broadcast reserved).

Private address ranges (RFC 1918) — not routable on the public Internet; used with NAT:

Class	Range	CIDR
A	10.0.0.0 – 10.255.255.255	10.0.0.0/8
B	172.16.0.0 – 172.31.255.255	172.16.0.0/12
C	192.168.0.0 – 192.168.255.255	192.168.0.0/16

(APIPA/link-local self-assigned range: 169.254.0.0/16 — appears when DHCP fails.)

8.10.2 Subnet mask, subnetting and CIDR

- **Subnet mask**: 32-bit pattern — 1s over the network+subnet part, 0s over the host part. `IP AND mask = network address`.
- **CIDR notation** `/n` = number of leading 1-bits: /24 ↔ 255.255.255.0, **/26 ↔ 255.255.255.192**, /30 ↔ 255.255.255.252.
- **Subnetting** borrows host bits to create subnets: borrowing s bits → **2^s subnets**, each with **2^(host bits remaining) − 2** hosts.
- **Block size trick:** block size = 256 − (mask's interesting octet); subnet boundaries step by the block size. E.g. mask .192 → block 64 → subnets start at 0, 64, 128, 192.
- CIDR (classless) also allows **supernetting/route aggregation** (e.g. combining four /24s into one /22).

(Worked Examples 4–7 below.)

8.10.3 IPv6 basics

Item	Detail
Address size	**128 bits** → $2^{128} \approx 3.4\times10^{38}$ addresses
Notation	Eight groups of 4 hex digits separated by colons: `2001:0db8:0000:0000:0000:ff00:0042:8329` ; leading zeros droppable; one run of zero groups → `::` (once only): `2001:db8::ff00:42:8329`
Loopback	`::1`
Header	Fixed **40 bytes**, simpler than IPv4's; extension headers chained
Removed from IPv4	Header checksum, router fragmentation, broadcast (replaced by multicast + **anycast**)
Built-in features	Auto-configuration (SLAAC), IPsec support, better QoS (flow label)
Transition	Dual stack, tunnelling (6to4), translation (NAT64)

8.10.4 TCP vs UDP ⭐

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	**Connection-oriented** (3-way handshake before data)	**Connectionless** (just send)
Reliability	Reliable — ACKs, retransmission, checksums	Unreliable, best-effort (checksum optional-ish)
Ordering	In-order delivery (sequence numbers)	No ordering guarantee
Flow / congestion control	Yes (sliding window / AIMD)	None
Speed / overhead	Slower; header **20–60 B**	Faster; header **8 B** (src port, dst port, length, checksum)
Data unit	Segment (byte-stream service)	Datagram (message service)
Used by	HTTP/HTTPS, FTP, SMTP, Telnet, SSH	DNS queries, DHCP, SNMP, TFTP, VoIP/streaming

TCP three-way handshake (connection establishment):

1. Client → Server: **SYN** (seq = x)
2. Server → Client: **SYN + ACK** (seq = y, ack = x+1)
3. Client → Server: **ACK** (ack = y+1) — connection established; data may flow.

Termination uses FIN/ACK exchanges (four-way). TCP header flags to know: SYN, ACK, FIN, RST, PSH, URG.

8.10.5 Port numbers ⭐ (memorise cold — direct one-mark questions)

Port = 16-bit number (0–65535) identifying a **process/application** on a host. Ranges: **well-known 0–1023**, registered 1024–49151, dynamic/ephemeral 49152–65535. Socket = IP address + port.

Protocol	Port	Transport	Purpose
FTP data / control	**20 / 21**	TCP	File transfer
SSH	**22**	TCP	Secure remote login
Telnet	**23**	TCP	Remote login (plaintext)
SMTP	**25**	TCP	Sending e-mail
DNS	**53**	UDP (TCP for zone transfer)	Name resolution
DHCP	**67 (server) / 68 (client)**	UDP	Auto IP configuration
TFTP	69	UDP	Trivial file transfer
HTTP	**80**	TCP	Web
POP3	**110**	TCP	Mail retrieval (download-and-delete)
NTP	123	UDP	Time sync
IMAP	**143**	TCP	Mail retrieval (server-side folders/sync)
SNMP agent / manager (trap)	**161 / 162**	UDP	Network management
HTTPS	**443**	TCP	Secure web (TLS)
RDP	3389	TCP	Remote desktop

8.10.6 Support protocols of the TCP/IP suite

Protocol	Layer	Function
ARP (Address Resolution Protocol)	between L2/L3	**IP address → MAC address**: sender broadcasts "who has IP x?"; owner replies with its MAC; result cached in ARP table
RARP	"	**MAC → IP** for diskless stations (obsolete; replaced by BOOTP → **DHCP**)
ICMP (Internet Control Message Protocol)	3	Error reporting & diagnostics for IP: destination unreachable, time exceeded (TTL=0), redirect, **echo request/reply — used by** `ping` ; `traceroute` exploits TTL-expiry messages
IGMP	3	Hosts join/leave multicast groups
DHCP	Application (UDP 67/68)	Dynamically leases IP address, mask, gateway, DNS to clients; **DORA** sequence: **D**iscover → **O**ffer → **R**equest → **A**cknowledge
DNS	Application (53)	**Domain name → IP address** (and reverse); distributed hierarchical database: root → TLD (.com/.in) → authoritative servers; record types A (IPv4), AAAA (IPv6), MX (mail), CNAME (alias)

8.10.7 Application-layer protocols

Protocol	Port	One-liner
HTTP	80	HyperText Transfer Protocol — web request/response (GET, POST…); **stateless**
HTTPS	443	HTTP over **TLS/SSL** — encrypted + server authenticated (padlock)
FTP	21 control + 20 data	File transfer using **two separate TCP connections** (control & data); active/passive modes; secure variants SFTP (over SSH), FTPS
SMTP	25	Simple Mail Transfer Protocol — **sending/relaying** mail (client→server, server→server); a **push** protocol
POP3	110	Post Office Protocol v3 — **downloads** mail to one device, typically deletes from server; simple
IMAP	143	Internet Message Access Protocol — mail stays **on the server**, folders synchronised across devices; more capable than POP3
Telnet	23	Remote terminal login — **everything in plaintext** including passwords → insecure
SSH	22	Secure Shell — **encrypted** remote login, command execution, tunnelling, SFTP/SCP; replaces Telnet

8.10.8 SNMP — Simple Network Management Protocol ⭐ (2025 PYQ)

Framework for **managing and monitoring devices** (routers, switches, servers, printers) on an IP network. Application-layer protocol running over **UDP: agent listens on port 161, manager receives traps on port 162**.

Model — Manager / Agent / MIB:

Element	Role
Manager (NMS)	Central management station software; polls agents, receives alarms, presents network status
Agent	Software on each managed device; maintains local management data and answers the manager
MIB (Management Information Base)	The hierarchical **database of managed objects/variables** on the agent (interfaces, counters, uptime…), each identified by an **OID**; structure defined by SMI (uses ASN.1)

Message types (direction matters — the exam point):

Message	Direction	Purpose
GetRequest (GET)	**Manager → Agent**	Read the value of a MIB variable
GetNextRequest	Manager → Agent	Read the next variable (walk the MIB tree)
GetBulkRequest (v2+)	Manager → Agent	Read large blocks efficiently
SetRequest (SET)	**Manager → Agent**	**Write/modify** a variable — i.e. configure/control the device
GetResponse (Response)	**Agent → Manager**	Reply carrying requested values / confirmation
Trap	**Agent → Manager**	**Unsolicited alert** on an event (link down, reboot, threshold crossed) — agent-initiated, no polling
Inform (v2+)	Manager/agent → manager	Acknowledged trap

> **PYQ takeaway:** *The manager issues GET (read) and SET (write) requests to the agent; the agent answers with Response and can asynchronously send TRAPs to the manager.* Any option saying the *agent* sends GET/SET to the manager, or that the manager sends traps, is wrong.

Versions: SNMP**v1** (basic, community-string "security"), SNMP**v2c** (adds GetBulk/Inform, better error handling, still community strings), SNMP**v3** (adds real **security: authentication, encryption/privacy, and access control**).

8.10.9 Internet concepts

Term	Meaning
Internet	Global public network of interconnected networks running TCP/IP
WWW	World Wide Web — the system of interlinked hypertext documents *on* the Internet (invented by **Tim Berners-Lee, 1989, CERN**); Web ≠ Internet — the Web is one service on it
URL	Uniform Resource Locator — address of a resource: `protocol://host[:port]/path?query` e.g. `https://www.rrb.gov.in:443/results/list.html`; parts = scheme, domain (host), optional port, path
ISP	Internet Service Provider — organisation providing Internet access (BSNL, Jio, Airtel); tiers 1/2/3
Browser / web server	Client that renders pages (Chrome/Firefox) / program serving them (Apache, Nginx, IIS)
Intranet	An organisation's **private** internal network using Internet technologies (TCP/IP, web) — accessible to members only
Extranet	Controlled extension of an intranet to **selected outsiders** (partners, vendors, customers) — "partly private"
VPN	Virtual Private Network — a **secure encrypted tunnel** over the public Internet linking remote users/sites to the private network as if locally connected; protocols: IPsec, SSL/TLS VPN, L2TP, (old) PPTP; gives confidentiality + site-to-site or remote-access connectivity cheaply vs leased lines
NAT	Network Address Translation — router rewrites **private** source IPs (RFC 1918) to its one **public** IP on the way out and maps replies back (PAT/NAPT uses port numbers to multiplex many hosts on one public address); conserves IPv4 addresses and hides internal topology

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8.11 Network Security (one page)

Goals — the CIA triad: Confidentiality (only authorised eyes), **I**ntegrity (no undetected alteration), **A**vailability (services up when needed); plus **authentication** (prove identity) and **non-repudiation** (sender can't deny sending — via digital signatures).

Firewalls — device/software enforcing an access policy between a trusted network and the outside:

Type	Works at	Idea
Packet-filtering	L3/L4	Permit/deny by source/dest IP, port, protocol — ACL rules; fast, stateless
Stateful inspection	L3/L4 + state	Tracks connections; allows only replies to legitimate outbound sessions
Application-level gateway (proxy)	L7	Understands the application protocol (HTTP, FTP); can filter content; terminates and re-originates connections
Circuit-level gateway	L5	Validates TCP handshakes/sessions (e.g. SOCKS)
DMZ	—	A buffer subnet between two firewalls hosting public-facing servers (web, mail)

Encryption:

- **Symmetric (secret-key):** the **same key** encrypts and decrypts — fast, suited to bulk data; problem = securely sharing the key. Algorithms: **DES (56-bit, broken), 3DES, AES (128/192/256 — current standard)**, RC4.
- **Asymmetric (public-key):** a mathematically linked **key pair** — encrypt with the receiver's **public** key, only the receiver's **private** key decrypts; also enables **digital signatures** (sign with private, verify with public). Algorithms: **RSA**, Diffie–Hellman (key exchange), ECC. Slower, so real systems (TLS/HTTPS) use asymmetric crypto to exchange a symmetric session key, then symmetric crypto for the data. **Hash functions** (MD5 — broken, SHA-256) provide integrity fingerprints.

Common threats (one-liners): Virus (attaches to files, needs host program), **worm** (self-replicates across the network on its own), **Trojan horse** (malicious code disguised as useful software), **spyware/keylogger**, **ransomware** (encrypts data for ransom), **phishing** (fraudulent mails/sites stealing credentials), **spoofing** (forged identity — IP/MAC/e-mail), **man-in-the-middle** (attacker secretly relays/alters traffic), **DoS/DDoS** (flooding a service to exhaust resources), **SQL injection** (malicious queries via input fields), **eavesdropping/sniffing** (packet capture on the medium). Defences: firewall + IDS/IPS, antivirus, patching, strong passwords/2-FA, encryption, backups.

8.12 Worked Examples (8)

Example 1 — Nyquist capacity.

A noiseless channel has a bandwidth of 3000 Hz and transmits a signal with 4 levels. Find the maximum bit rate. What if 8 levels are used?

$C = 2\,B \log_2 L = 2 \times 3000 \times \log_2 4 = 6000 \times 2 =$ **12,000 bps**.

With $L = 8$: $C = 2 \times 3000 \times 3 =$ **18,000 bps**. (Each extra bit per level adds $2B = 6000$ bps.)

Example 2 — Shannon capacity (telephone line).

A telephone channel has bandwidth 3000 Hz (300–3300 Hz) and SNR = 3162. Find the channel capacity. Also express the SNR in dB.

$C = B \log_2(1 + \text{SNR}) = 3000 \times \log_2(3163) \approx 3000 \times 11.63 \approx$ **34,880 bps** (~34.9 kbps — the theoretical basis of the ~33.6 kbps analog modem limit).

$\text{SNR}_{dB} = 10 \log_{10} 3162 \approx 10 \times 3.5 =$ **35 dB**.

Example 3 — Shannon then Nyquist (combined design).

A channel has B = 1 MHz and SNR = 63. (a) Find the upper limit on data rate. (b) If we operate at 4 Mbps, how many signal levels are needed?

(a) $C = 10^6 \times \log_2(1 + 63) = 10^6 \times \log_2 64 = 10^6 \times 6 =$ **6 Mbps**.

(b) Nyquist: $4\times10^6 = 2 \times 10^6 \times \log_2 L \rightarrow \log_2 L = 2 \rightarrow$ **L = 4 levels**. (4 Mbps < 6 Mbps, so achievable.)

Example 4 — Address class identification.

Identify the class of: (i) 10.34.2.1 (ii) 130.45.6.7 (iii) 200.11.5.9 (iv) 229.1.1.1 (v) 248.0.0.1.

Look at the first octet: (i) 10 ∈ 1–126 → **Class A (also a private address)**; (ii) 130 ∈ 128–191 → **Class B**; (iii) 200 ∈ 192–223 → **Class C**; (iv) 229 ∈ 224–239 → **Class D (multicast)**; (v) 248 ∈ 240–255 → **Class E (experimental)**.

Example 5 — Network address and host count.

A host has IP 192.168.10.77 with subnet mask 255.255.255.0. Find the network address, broadcast address, and number of usable hosts.

Mask /24 → network = first three octets: **192.168.10.0**. Broadcast = host bits all 1: **192.168.10.255**. Host bits = 8 → usable hosts = $2^8 - 2 =$ **254** (range 192.168.10.1 – .254).

Example 6 — Subnetting a Class C network.

Divide 192.168.1.0/24 into subnets that support at least 50 hosts each. Give the mask, number of subnets, and the subnet address of host 192.168.1.130.

Need ≥ 50 hosts → host bits h with $2^h - 2 \geq 50 \rightarrow h = 6$ (62 hosts). So mask = 32 − 6 = **/26 = 255.255.255.192**. Borrowed bits = 2 → **4 subnets**, block size 256 − 192 = 64: subnets **.0, .64, .128, .192**.

130 lies in 128–191 → host 192.168.1.130 is in subnet **192.168.1.128/26** (broadcast 192.168.1.191, usable .129–.190).

Example 7 — CIDR block reading.

For 172.16.40.20/12: state the mask in dotted decimal and the number of addresses in the block.

/12 → 12 ones: 255.240.0.0 (**255.11110000.0.0**). Host bits = 32 − 12 = 20 → addresses = $2^{20} =$ **1,048,576** (usable hosts $2^{20} - 2 =$ 1,048,574). Note 172.16.0.0/12 is exactly the RFC 1918 private Class-B block (172.16.0.0 – 172.31.255.255).

Example 8 — Bit rate vs baud rate.

A modem uses 8-PSK (8 phase states) and signals at 1600 baud. Find the bit rate. Conversely, what baud rate carries 72,000 bps using 64-QAM?

8-PSK → $\log_2 8 = 3$ bits/symbol → bit rate = 1600 × 3 = **4800 bps**.

64-QAM → 6 bits/symbol → baud = 72,000 / 6 = **12,000 baud**.

8.13 PYQ Corner — actual 22-04-2025 CBT-2 questions, solved

PYQ 1. Which of the following statements correctly differentiates a modem from a Network Interface Card (NIC)?

(a) A NIC converts digital signals to analog for transmission over telephone lines, while a modem connects a computer to the LAN

(b) **A modem converts digital signals to analog (and analog back to digital) for transmission over telephone lines, while a NIC provides the interface that connects a computer to a network and carries a unique MAC address**

(c) Both perform identical functions and are interchangeable

(d) A modem operates at the data-link layer while a NIC operates at the application layer

Answer: (b). The modem's defining job is signal conversion (MOdulation–DEModulation) so digital data can ride an analog channel; the NIC is the host's Layer-1/2 network attachment hardware with its burned-in 48-bit MAC address. (a) swaps the roles; (c) is plainly wrong; (d) misplaces both layers. *(See table §8.3.4.)*

PYQ 2. Which statement about twisted-pair cables is INCORRECT?

(a) STP contains a metal shield around the pairs that reduces electromagnetic interference

(b) UTP is cheaper and easier to install than STP

(c) **UTP offers higher resistance to electromagnetic interference than STP**

(d) Twisting of the wires helps cancel externally induced noise

Answer: (c) — this is the false statement. The grounded metallic shield of **STP** is precisely what gives it **better** EMI immunity; UTP relies on twisting alone, so its EMI resistance is *lower*. Statements (a), (b), (d) are all true. RRB framed this as "choose the incorrect statement," so the option praising UTP's EMI resistance is the answer. *(See §8.2.1.)*

PYQ 3. In which network topology does the failure of the central device bring down the entire network, while the failure of any single node leaves the rest of the network unaffected?

(a) Bus (b) Ring (c) Mesh (d) **Star**

Answer: (d) Star. In star topology each node has its own dedicated link to a central hub/switch: a node or its cable failing isolates only that node, but the central device is a single point of failure. Contrast: **bus** — backbone break kills all; **ring** — any single node/link break can stop the whole (single) ring; **mesh** — survives individual failures because traffic reroutes over alternative links. *(Master table §8.5.3.)*

PYQ 4. In the SNMP framework, which statement about manager–agent communication is CORRECT?

(a) The agent sends GET and SET requests to the manager to retrieve configuration data

(b) **The manager sends GET requests to read MIB variables from the agent and SET requests to modify them; the agent replies with responses and can send unsolicited TRAP messages to the manager**

(c) TRAP messages are sent by the manager to the agent to poll device status

(d) The MIB is maintained by the manager, and agents query it using GetNext

Answer: (b). SNMP direction rules: **GET/GetNext/GetBulk/SET flow manager → agent** (read/write of MIB objects); **Response and TRAP flow agent → manager** (Trap = asynchronous, event-driven alert, no request needed). The **MIB resides on the agent** (managed device), not the manager. Ports: agent 161, manager trap listener 162, over UDP. *(See §8.10.8.)*

8.14 Practice MCQs (25)

Q1. The transmission mode in which data flows in both directions but only one direction at a time is

(a) simplex (b) half-duplex (c) full-duplex (d) multiplex

Q2. In asynchronous serial transmission, each character is framed by

(a) a preamble and CRC (b) a start bit and stop bit(s) (c) sync bytes only (d) flag bytes 01111110

Q3. A signal uses 16 levels (16-QAM) at 3000 baud. The bit rate is

(a) 3000 bps (b) 6000 bps (c) 12,000 bps (d) 48,000 bps

Q4. The maximum data rate of a noiseless 4 kHz channel using binary (2-level) signalling is

(a) 4 kbps (b) 8 kbps (c) 16 kbps (d) 2 kbps

Q5. A channel of bandwidth 20 kHz has SNR = 1023. Its Shannon capacity is

(a) 100 kbps (b) 200 kbps (c) 20 kbps (d) 400 kbps

Q6. Which cable category supports 1 Gbps Ethernet over 100 m at 100 MHz bandwidth?

(a) Cat 3 (b) Cat 5e (c) Cat 7 only (d) RG-58

Q7. Optical fibre guides light by the principle of

(a) refraction through the cladding (b) total internal reflection (c) diffraction (d) polarisation

Q8. Which transmission medium is completely immune to electromagnetic interference?

(a) UTP (b) STP (c) coaxial cable (d) optical fibre

Q9. In Manchester encoding, the transition in the middle of each bit interval provides

(a) error correction (b) compression (c) self-clocking/synchronisation (d) encryption

Q10. The 48-bit MAC address of a NIC — the first three bytes represent

(a) the host serial number (b) the OUI (manufacturer ID) (c) the IP network (d) the port number

Q11. The RS-232 standard represents logic 1 by a voltage of

(a) +3 to +15 V (b) 0 V (c) −3 to −15 V (d) +5 V

Q12. A full mesh network of 10 nodes requires how many duplex links?

(a) 10 (b) 20 (c) 45 (d) 90

Q13. In which topology does a break anywhere in the single shared backbone cable stop all communication?

(a) star (b) bus (c) mesh (d) tree

Q14. Which OSI layer is responsible for routing and logical (IP) addressing?

(a) data link (b) transport (c) network (d) session

Q15. The PDU at the data-link layer is called a

(a) segment (b) packet (c) frame (d) bit

Q16. Encryption and data compression are functions of which OSI layer?

(a) session (b) presentation (c) application (d) transport

Q17. A switch differs from a hub mainly because a switch

(a) works at the physical layer (b) broadcasts every frame to all ports (c) forwards frames only to the destination port using a MAC table (d) converts digital signals to analog

Q18. Which device separates broadcast domains by default?

(a) repeater (b) switch (c) bridge (d) router

Q19. In CSMA/CD, after detecting a collision a station transmits a

(a) token (b) beacon (c) jam signal (d) preamble

Q20. The minimum Ethernet frame length (excluding preamble) is

(a) 46 bytes (b) 64 bytes (c) 128 bytes (d) 1518 bytes

Q21. IEEE 802.11n differs from 802.11g chiefly by introducing

(a) infrared links (b) MIMO and operation in both 2.4 and 5 GHz (c) token passing (d) 53-byte cells

Q22. The IP address 172.20.5.9 belongs to

(a) Class A public (b) Class B private range (c) Class C private range (d) Class D multicast

Q23. The subnet mask 255.255.255.240 in CIDR notation is /28. The number of usable hosts per subnet is

(a) 16 (b) 14 (c) 30 (d) 12

Q24. Which sequence is the correct TCP three-way handshake?

(a) ACK, SYN, SYN-ACK (b) SYN, SYN-ACK, ACK (c) SYN, ACK, FIN (d) FIN, ACK, SYN

Q25. Default port numbers of HTTPS, DNS and SMTP respectively are

(a) 80, 25, 53 (b) 443, 53, 25 (c) 443, 25, 110 (d) 22, 53, 25

Answer key with brief explanations

Q	Ans	Reason
1	b	Half-duplex = both directions, one at a time (walkie-talkie)
2	b	Async framing = start bit (0) + data + stop bit(s) (1)
3	c	16 levels → 4 bits/symbol; 3000 × 4 = 12,000 bps
4	b	Nyquist: $2 \times 4000 \times \log_2 2 = 8000$ bps
5	b	$C = 20k \times \log_2 1024 = 20k \times 10 = 200$ kbps
6	b	Cat 5e: 1 Gbps at 100 MHz (Cat 7 also does but 600 MHz/shielded — "only" makes c wrong)
7	b	Core n > cladding n → total internal reflection
8	d	Fibre carries light — no electrical pickup at all
9	c	Guaranteed mid-bit transition = embedded clock
10	b	First 3 bytes = IEEE-assigned OUI; last 3 = device serial
11	c	RS-232 is negative logic: mark (1) = −3…−15 V
12	c	$n(n-1)/2 = 10 \times 9/2 = 45$
13	b	Bus backbone is the single shared path; break kills the LAN
14	c	Network layer = logical addressing + routing
15	c	Bits→(L2) frames→(L3) packets→(L4) segments
16	b	Presentation = translation, encryption, compression
17	c	Switch learns MACs (CAM table) and unicasts frames
18	d	Bridges/switches split collision domains; only a router (or VLAN) splits broadcast domains
19	c	32-bit jam signal ensures every station sees the collision
20	b	64 B minimum (slot-time requirement); payload padded to ≥ 46 B
21	b	802.11n = Wi-Fi 4: MIMO, 40 MHz channels, dual-band, up to 600 Mbps
22	b	172.16.0.0–172.31.255.255 is the Class-B private (RFC 1918) block; 172.20 falls inside
23	b	/28 → 4 host bits → $2^4 - 2 = 14$ usable

Q	Ans	Reason
24	b	SYN → SYN+ACK → ACK
25	b	HTTPS 443, DNS 53, SMTP 25

8.15 Rapid-Revision Sheet

OSI at a glance

L	Layer	Keyword	PDU	Address	Device
7	Application	User services	Data	—	Gateway/firewall(L7)
6	Presentation	Translate, **encrypt, compress**	Data	—	—
5	Session	Dialog, **checkpoints**	Data	—	—
4	Transport	**Process-to-process**, TCP/UDP	Segment	**Port**	Firewall (L4)
3	Network	**Routing**, host-to-host	Packet	**IP**	**Router**
2	Data Link	Framing, hop-to-hop, CRC	Frame	**MAC**	**Switch, bridge, NIC**
1	Physical	Raw bits, media	Bit	—	**Hub, repeater, modem, cables**

Mnemonic (bottom-up): **P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way.

Port numbers (cold-memory list)

FTP **20/21** · SSH **22** · Telnet **23** · SMTP **25** · DNS **53** · DHCP **67/68** · TFTP 69 · HTTP **80** · POP3 **110** · NTP 123 · IMAP **143** · **SNMP 161/162** · HTTPS **443** · RDP 3389.

Cable categories

Cat3 = 10 Mbps · **Cat5e = 1 Gbps/100 MHz** · **Cat6 = 1 G (10 G @55 m)/250 MHz** · Cat6a = 10 G/500 MHz · **Cat7 = 10 G/600 MHz, shielded** · UTP segment max **100 m** · connector **RJ-45** · **STP beats UTP on EMI; UTP wins on cost/ease**.

Device vs layer

Hub/repeater/modem → **L1** · NIC → L1–2 · bridge/switch/AP → **L2** · router → **L3** · gateway → up to **L7** · switch splits **collision** domains · router splits **broadcast** domains.

Topology snapshot

	Bus	Star	Ring	Mesh
Cable cost	least	medium	medium	most (n(n−1)/2)
Node fails	OK	**OK (only that node)**	ring breaks	OK (reroute)
Cable/backbone fails	**all down**	one node down	all down	OK (reroute)
Single point of failure	backbone	**hub/switch**	every node/link	none
Access	CSMA/CD	switched	token	routed

Numbers to remember

- Nyquist **$C = 2B \log_2 L$**; Shannon **$C = B \log_2(1+SNR)$**, SNR linear.
- Bit rate = baud × $\log_2$(levels).
- MAC = **48 bits**; IPv4 = **32 bits**; IPv6 = **128 bits**; port = 16 bits.
- Class ranges (1st octet): A 1–126, B 128–191, C 192–223, D 224–239 (multicast), E 240–255; loopback **127.0.0.1**.
- Private: **10/8, 172.16/12, 192.168/16**; APIPA 169.254/16.
- Hosts per subnet = 2^h − 2; subnets from s borrowed bits = 2^s; block size = 256 − mask octet.
- Ethernet frame **64–1518 B**; payload 46–1500 B; CRC-32; jam = 32 bits; min frame ↔ slot time.
- TCP header 20–60 B (reliable, ordered, 3-way SYN→SYN-ACK→ACK); UDP header **8 B** (fast, connectionless).
- DHCP = **DORA**; ARP: IP→MAC (broadcast request, unicast reply); ICMP → ping/traceroute; DNS: name→IP.
- SNMP: manager **GET/SET** → agent; agent **Response/TRAP** → manager; MIB lives **on the agent**; UDP **161/162**; v3 adds security.
- GEO satellite 35,786 km, ~270 ms hop delay; ISDN BRI = 2B+D = 144 kbps; ATM cell = **53 B (48+5)**.
- 802.11: a/g = 54 M, b = 11 M, n = 600 M (MIMO), ac = 5 GHz gigabit, ax = Wi-Fi 6.

End of Chapter 8. Next: Chapter 9 — Computer Programming (C/C++, Data Structures, DBMS).